

```
1 %% function f_find_spherepos
2 function [spherePos] = f_find_spherepos( Data, Config, echogram, ✓
range, approxDist, rangebuffer)
3 %% function [spherePos] = f_find_spherepos( echogram, range, ✓
approxDist, rangebuffer)
4     range = range(1,:);
5     Nsamp = size(echogram,1); % Number of samples == number of pings
6
7     if isempty(rangebuffer)
8         rangebuffer = 0.5; % m
9     end
10
11 % %     spherePos.index = zeros(Nsamp,1);
12 % %     spherePos.range = zeros(Nsamp,1);
13
14 % ✓
----- ✓
-
15     %% Find range, index, and amplitude of sphere target
16     % ✓
----- ✓
-
17     f_disp_dbstack_info()
18     fprintf('Find range, index, and amplitude of sphere target.\n');
19
20     approxDist = approxDist;
21     i=find((range>approxDist-rangebuffer) & ✓
(range<approxDist+rangebuffer));
22 %     if(length(i)>1)
23 %         i = i(1);
24 %     end
25     spherePos.index = zeros(Nsamp,1);
26     spherePos.indexWithTx = zeros(Nsamp,1); % account for Ltx = size ✓
(Data.RawEchol_90kHzTx_DataI,2);
27     spherePos.range = zeros(Nsamp,1);
28
29 %     % TO DO - MAKE THIS SAFE:
30 %     Ltx = size(Data.RawEchol_90kHzTx_DataI,2); % Data. ✓
RawEchol_90kHzTx_DataI MUST EXIST
31
32     for k=1:Nsamp
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33     indc = i( find(echogram(k,i) == max((echogram(k,i)))) );
34     %spherePos(k) = r(j);
35     spherePos.index(k,1) = indc;
36     %spherePos.indexWithTx(k,1) = indc+Ltx;
37     spherePos.indexWithTx(k,1) = -99999;
38     spherePos.range(k,1) = range(indc);
39     spherePos.ampLin(k,1) = echogram(k,indc);
40 end
41
42 spherePos.indexmean = nanmean( spherePos.index );
43 spherePos.indexmeanWithTx = nanmean( spherePos.indexWithTx );
44 spherePos.ampLinmean = nanmean( spherePos.ampLin );
45 spherePos.rangemean = nanmean( spherePos.range );
46 spherePos.ampLog = 10*log10( spherePos.ampLin );
47 spherePos.ampLogmean = 10*log10( spherePos.ampLinmean );
48
49 fprintf('nanmean(spherePos.index) = %.1f (sample)\n', nanmean(
(spherePos.index) );
50 fprintf('nanmean(spherePos.indexWithTx) = %.1f (sample)\n',
nanmean(spherePos.indexWithTx) );
51 fprintf('nanmean(spherePos.range) = %.2f (m)\n', nanmean(
(spherePos.range) );
52 end
53
```